

Statistics & Probability



Mean, median, and mode

- 1 Find the mean of 4, 8, 6, 10, 7.
 - 2 Find the median of 3, 9, 5, 1, 7.
 - 3 Find the mode of 2, 4, 4, 6, 8, 4, 2.
 - 4 A data set has values 8, 12, 10, 15, 10, 14, 11. Find the mean, median, and mode.
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Range and quartiles

- 5 Find the range of 5, 12, 8, 20, 3.
 - 6 For the data set 2, 4, 5, 7, 8, 9, 12, find Q_1 and Q_3 .
 - 7 Find the interquartile range (IQR) for the same data set.
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Variance and standard deviation

- 8 Find the variance of 2, 4, 6, 8, 10, using $\sigma^2 = \frac{\sum (x - \mu)^2}{n}$.
 - 9 Find the standard deviation of the same data set, rounded to two decimal places.
 - 10 A data set has variance 25. State its standard deviation.
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Frequency tables

- 11 A frequency table shows value 1 with frequency 3, value 2 with frequency 5, and value 3 with frequency 2. Find the mean.
 - 12 Using the same frequency table, find the mode.
 - 13 Using the same frequency table, find the median.
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Basic probability

- 14 A fair six-sided die is rolled. Find $P(\text{number} > 4)$.
- 15 A bag contains 5 red and 3 blue marbles. Find $P(\text{blue})$.
- 16 A card is drawn from a standard 52-card deck. Find $P(\text{king})$.
- 17 Two fair coins are flipped. Find $P(\text{both heads})$.
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Complementary and mutually exclusive events

- 18 If $P(A) = 0.35$, find $P(\text{not } A)$.
- 19 A die is rolled. Events $A = \{\text{even}\}$ and $B = \{\text{rolling a } 5\}$. Are A and B mutually exclusive?
- 20 If $P(A) = 0.4$ and $P(B) = 0.25$, and A, B are mutually exclusive, find $P(A \text{ or } B)$.
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Independent events

- 21 Two independent events have $P(A) = 0.6$ and $P(B) = 0.5$. Find $P(A \text{ and } B)$.
- 22 A fair coin is flipped twice. Find $P(\text{first heads and second tails})$.
- 23 A card is drawn from a deck, replaced, then a second card is drawn. Find $P(\text{both are aces})$.
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Conditional probability

- 24 In a class, $P(\text{math}) = 0.7$ and $P(\text{physics and math}) = 0.4$. Find $P(\text{physics} \mid \text{math})$.
- 25 A box has 10 balls: 6 red and 4 blue. Two balls are drawn without replacement. Find $P(\text{2nd is blue} \mid \text{1st is blue})$.
- 26 Given $P(A) = 0.5$, $P(B) = 0.3$, $P(A \text{ and } B) = 0.15$, find $P(A \mid B)$.
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Counting: permutations and combinations

- 27 In how many ways can 4 books be arranged on a shelf?

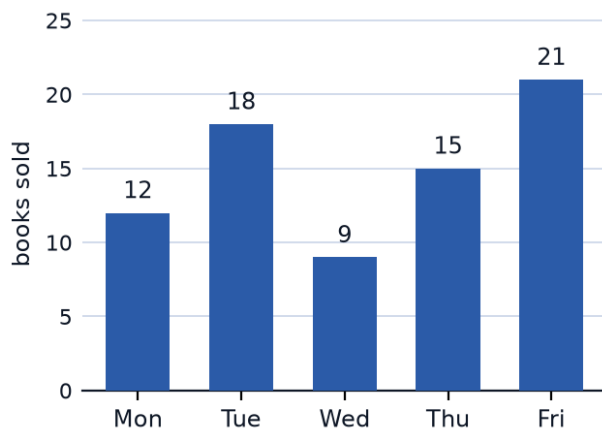
- 28 How many ways can a committee of 3 be chosen from 8 people (order doesn't matter)?
- 29 A code uses 3 distinct digits from 0–9, where order matters. How many possible codes are there?
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The five-number summary

- 30 For the data set 6, 10, 14, 18, 22, 26, 30, find the five-number summary (minimum, Q_1 , median, Q_3 , maximum).
- 31 Find the IQR for the same data set.
- 32 Outlier boundaries are $Q_1 - 1.5 \times \text{IQR}$ and $Q_3 + 1.5 \times \text{IQR}$. Find both boundaries for the same data set.
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Reading data from a bar chart

- 33 The bar chart shows the number of books sold each day at a bookstore over five days.



- a Find the total number of books sold over the five days.
- b Find the mean number of books sold per day.
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Word problems: statistics and probability in context

- 34 This question has two parts. A survey of 200 students found that 120 like math and 90 like science, while 50 like both subjects.
- a Find the probability that a randomly selected student likes math or science, using $P(M \cup S) = P(M) + P(S) - P(M \cap S)$.
- b Find the probability that a student likes neither subject.

- 35 This question has two parts. A quality-control inspector finds that 2% of items from a production line are defective. A random sample of 3 items is tested, and defects occur independently.
- a Find the probability that all three items are defective.
 - b Find the probability that none of the three items are defective.
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MCQ — Mean, median, and mode

- 36 Find the mean of 5, 9, 7, 11, 8.
- a. 8
 - b. 7
 - c. 40
 - d. 9
- 37 Find the median of 4, 10, 6, 2, 8.
- a. 5
 - b. 8
 - c. 6
 - d. 4
- 38 Find the mode of 3, 5, 5, 7, 9, 5, 3.
- a. 5
 - b. 4
 - c. 3
 - d. 7

- 39 A data set has values 9, 13, 11, 16, 11, 15, 12. Find the mean, median, and mode.
- a. Mean = 13, median = 12, mode = 11
 - b. Mean = 12.43, median = 11, mode = 12
 - c. Mean $\approx$ 12.43, median = 12, mode = 11
 - d. Mean $\approx$ 12.43, median = 13, mode = 11
- 40 Find the mean of 12, 18, 15, 21, 14.
- a. 15
 - b. 16
 - c. 80
 - d. 18
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MCQ — Range and quartiles

- 41 Find the range of 6, 14, 9, 22, 4.
- a. 22
 - b. 16
 - c. 18
 - d. 20
- 42 For the data set 3, 5, 6, 8, 9, 10, 13, find Q_1 and Q_3 .
- a. $Q_1 = 5$, $Q_3 = 9$
 - b. $Q_1 = 5$, $Q_3 = 10$
 - c. $Q_1 = 3$, $Q_3 = 13$
 - d. $Q_1 = 6$, $Q_3 = 9$

43 Find the interquartile range (IQR) for the same data set.

- a. 10
- b. 8
- c. 3
- d. 5

44 Find the range of 18, 25, 12, 30, 9.

- a. 18
- b. 16
- c. 21
- d. 30

MCQ — Variance and standard deviation

45 Find the variance of 3, 5, 7, 9, 11, using $\sigma^2 = \frac{\sum (x - \mu)^2}{n}$.

- a. 4
- b. 2.83
- c. 8
- d. 40

46 Find the standard deviation of the same data set, rounded to two decimal places.

- a. 1.41
- b. 4.00
- c. 2.83
- d. 8.00

47 A data set has variance 36. State its standard deviation.

- a. 6
- b. 12
- c. 18
- d. 1,296

48 Find the variance of 4, 8, 12, 16, 20, using $\sigma^2 = \frac{\sum(x - \mu)^2}{n}$.

- a. 160
- b. 5.66
- c. 16
- d. 32

MCQ — Frequency tables

49 A frequency table shows value 1 with frequency 4, value 2 with frequency 6, and value 3 with frequency 5. Find the total number of data points.

- a. 10
- b. 15
- c. 3
- d. 6

50 Using the same frequency table, find the mode.

- a. 6
- b. 3
- c. 2
- d. 1

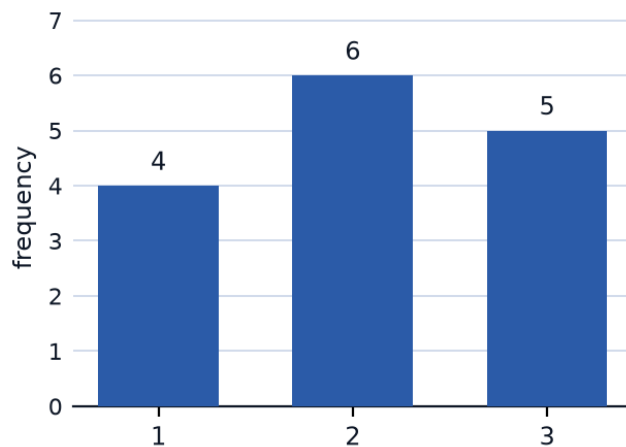
51 Using the same frequency table, find the median.

- a. 1.5
- b. 1
- c. 2
- d. 3

52 Using the same frequency table, find the mean.

- a. 2.33
- b. ≈ 2.07
- c. 2.00
- d. 1.87

53 The bar chart shows a frequency table. Find the total number of data points.



- a. 15
- b. 6
- c. 3
- d. 10

MCQ — Basic probability

54 A fair six-sided die is rolled. Find $P(\text{number} < 3)$.

a. $\frac{1}{6}$

b. $\frac{1}{2}$

c. $\frac{1}{3}$

d. $\frac{2}{3}$

55 A bag contains 7 red and 4 blue marbles. Find $P(\text{red})$.

a. $\frac{7}{11}$

b. $\frac{1}{11}$

c. $\frac{4}{11}$

d. $\frac{7}{4}$

56 A card is drawn from a standard 52-card deck. Find $P(\text{queen})$.

a. $\frac{4}{13}$

b. $\frac{1}{4}$

c. $\frac{1}{13}$

d. $\frac{1}{52}$

57 Two fair coins are flipped. Find $P(\text{at least one tail})$.

a. $\frac{1}{2}$

b. $\frac{3}{4}$

c. $\frac{1}{4}$

d. $\frac{1}{3}$

58 A fair six-sided die is rolled. Find $P(\text{multiple of } 3)$.

- a. $\frac{2}{5}$
- b. $\frac{1}{3}$
- c. $\frac{1}{2}$
- d. $\frac{1}{6}$

MCQ — Complementary and mutually exclusive events

59 If $P(A) = 0.28$, find $P(\text{not } A)$.

- a. 0.28
- b. 1.28
- c. 0.5
- d. 0.72

60 A die is rolled. Events $A = \{\text{odd}\}$ and $B = \{\text{rolling a } 4\}$. Are A and B mutually exclusive?

- a. Yes
- b. Only sometimes
- c. No
- d. Cannot be determined

61 If $P(A) = 0.3$ and $P(B) = 0.45$, and A, B are mutually exclusive, find $P(A \text{ or } B)$.

- a. 0.75
- b. 0.15
- c. 1.05
- d. 0.135

62 If $P(A) = 0.6$, find $P(\text{not } A)$.

- a. 1.6
- b. 0.4
- c. 0.5
- d. 0.6

MCQ — Independent events

63 Two independent events have $P(A) = 0.7$ and $P(B) = 0.4$. Find $P(A \text{ and } B)$.

- a. 1.1
- b. 0.28
- c. 0.3
- d. 0.175

64 A fair coin is flipped twice. Find $P(\text{both heads})$.

- a. $\frac{1}{3}$
- b. 1
- c. $\frac{1}{4}$
- d. $\frac{1}{2}$

65 A card is drawn from a deck, replaced, then a second card is drawn. Find $P(\text{both are kings})$.

- a. $\frac{2}{13}$
- b. $\frac{1}{26}$
- c. $\frac{1}{13}$
- d. $\frac{1}{169}$

- 66 Two independent events have $P(A) = 0.5$ and $P(B) = 0.8$. Find $P(A \text{ and } B)$.
- a. 0.65
 - b. 1.3
 - c. 0.3
 - d. 0.4
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MCQ — Conditional probability

- 67 In a class, $P(\text{science}) = 0.65$ and $P(\text{math and science}) = 0.3$. Find $P(\text{math} \mid \text{science})$.
- a. 0.65
 - b. 0.195
 - c. ≈ 0.46
 - d. 0.3
- 68 A box has 12 balls: 7 red and 5 blue. Two balls are drawn without replacement. Given the first is blue, find $P(\text{second is blue})$.
- a. $\frac{5}{11}$
 - b. $\frac{4}{11}$
 - c. $\frac{5}{12}$
 - d. $\frac{4}{12}$
- 69 Given $P(A) = 0.6$, $P(B) = 0.4$, $P(A \text{ and } B) = 0.2$, find $P(A \mid B)$.
- a. 0.3
 - b. 0.33
 - c. 0.8
 - d. 0.5

70 Given $P(A) = 0.45$, $P(B) = 0.5$, $P(A \text{ and } B) = 0.2$, find $P(B | A)$.

- a. 0.4
- b. 0.09
- c. 0.5
- d. ≈ 0.44

MCQ — Counting: permutations and combinations

71 In how many ways can 5 books be arranged on a shelf?

- a. 25
- b. 120
- c. 20
- d. 60

72 How many ways can a committee of 4 be chosen from 9 people (order doesn't matter)?

- a. 126
- b. 3,024
- c. 362,880
- d. 36

73 A code uses 4 distinct digits from 0–9, where order matters. How many possible codes are there?

- a. 10,000
- b. 40
- c. 210
- d. 5,040

74 In how many ways can 6 people be arranged in a line?

- a. 720
- b. 120
- c. 36
- d. 6

MCQ — The five-number summary

75 For the data set 5, 9, 13, 17, 21, 25, 29, find the five-number summary (minimum, Q_1 , median, Q_3 , maximum).

- a. 9, 13, 17, 21, 25
- b. 5, 9, 17, 25, 29
- c. 5, 13, 17, 21, 29
- d. 5, 9, 13, 25, 29

76 Find the IQR for the same data set.

- a. 8
- b. 24
- c. 16
- d. 20

77 Outlier boundaries are $Q_1 - 1.5 \times IQR$ and $Q_3 + 1.5 \times IQR$. Find both boundaries for the same data set.

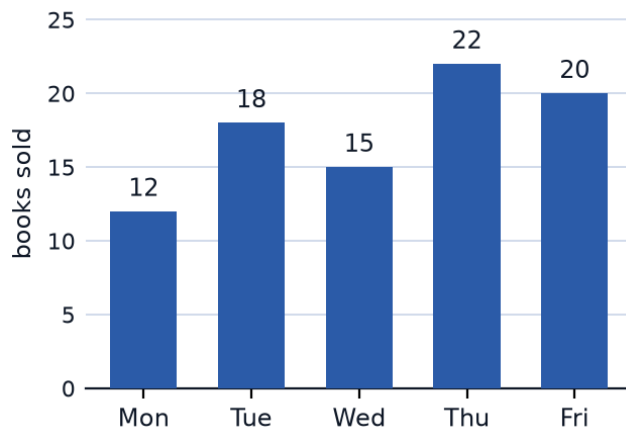
- a. -15 and 49
- b. 9 and 25
- c. -15 and 40
- d. -24 and 40

78 For the data set 10, 15, 20, 25, 30, 35, 40, find the median.

- a. 25
- b. 30
- c. 20
- d. 27.5

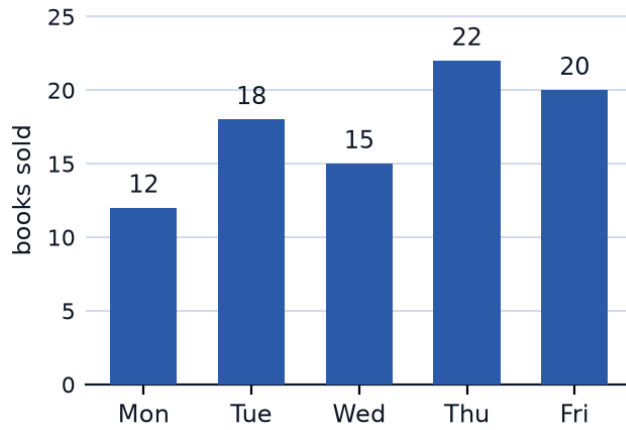
MCQ — Reading data from a bar chart

79 The bar chart shows the number of books sold each day at a bookstore over five days. Find the total books sold.



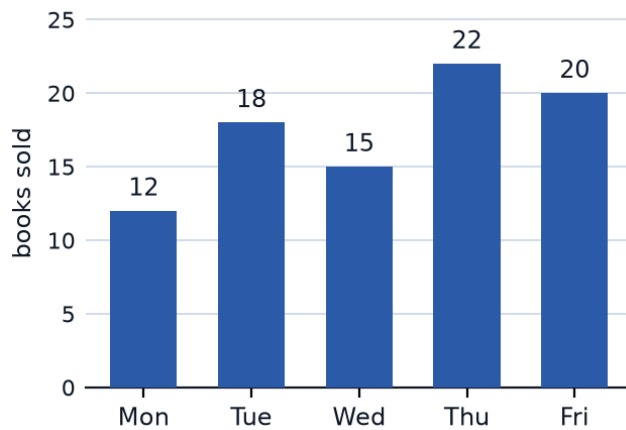
- a. 87
- b. 22
- c. 77
- d. 97

80 Using the same bar chart, find the mean number of books sold per day.



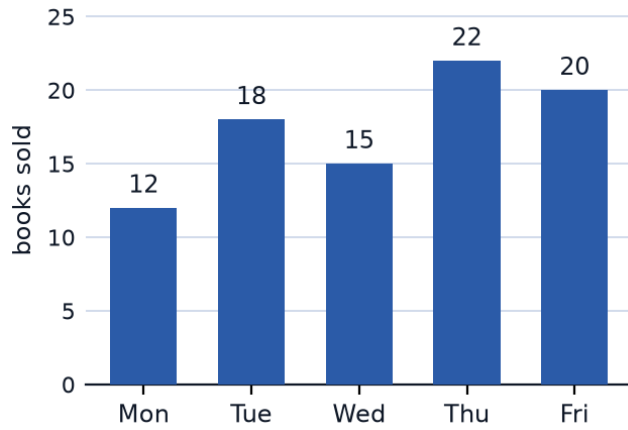
- a. 15
- b. 18
- c. 20
- d. 17.4

81 Using the same bar chart, find the range of books sold.



- a. 12
- b. 22
- c. 8
- d. 10

82 Using the same bar chart, on which day were the most books sold?



- a. Wednesday
- b. Thursday
- c. Tuesday
- d. Friday

MCQ — Word problems: statistics and probability in context

83 A survey of 250 students found that 150 like math and 100 like science, with 40 liking both. Find the probability that a randomly selected student likes math or science.

- a. 0.84
- b. 0.6
- c. 0.16
- d. 1.0

- 84** A quality-control inspector finds that 3% of items from a production line are defective. If 500 items are inspected, how many are expected to be defective?
- a.** 150
 - b.** 15
 - c.** 3
 - d.** 30
- 85** In a survey of 180 people, 108 prefer coffee over tea. Find the probability, as a percent, that a randomly selected person prefers coffee.
- a.** 60%
 - b.** 50%
 - c.** 108%
 - d.** 40%